

1 Cross Norms

Notation 1.1. In this section, A_1, A_2 will denote Banach spaces. The norms on these spaces will be denoted by $\|\cdot\|$ unambiguously as we will distinguish between elements of A_1 and A_2 .

Notation 1.2. For Banach spaces A_1, A_2 , we will denote their *algebraic tensor product* by $A_1 \otimes A_2$. Its completion in some norm $\|\cdot\|_\beta$ will be denoted by $A_1 \otimes_\beta A_2$.

Definition 1.3. A norm $\|\cdot\|_c$ on $A_1 \otimes A_2$ is called a *cross norm* if it satisfies $\|x_1 \otimes x_2\|_c = \|x_1\| \|x_2\|$ for all $x_1 \in A_1, x_2 \in A_2$.

Definition 1.4. The *injective cross norm* $\|\cdot\|_\lambda$ on $A_1 \otimes A_2$ is defined by

$$\|\xi\| = \sup\left\{\left|\sum_{k=1}^n f(x_{1,k})g(x_{2,k})\right| : \begin{array}{l} \xi = \sum_{k=1}^n x_{1,k} \otimes x_{2,k} = \xi, \\ f \in A_1^*, \|f\| = 1, \\ g \in A_2^*, \|g\| = 1 \end{array} \right\}.$$

Remark 1.5. That $\|\cdot\|_\lambda$ is a cross norm is immediate from an application of the Hahn Banach Theorem.

Remark 1.6. Suppose a norm $\|\cdot\|_\beta$ satisfies $\|f \otimes g\|_{\beta^*} \leq \|f\| \|g\|$, then we also have

$$\left|\sum_{j=1}^m f(x_{1,j})g(x_{2,j})\right| \leq \|f \otimes g\|_{\beta^*} \left\|\sum_{j=1}^m x_{1,k} \otimes x_{2,j}\right\|_\beta \leq \left\|\sum_{j=1}^m x_{1,k} \otimes x_{2,j}\right\|_\beta$$

for f in A_1^* with $\|f\| \leq 1$ and $g \in A_2^*$ with $\|g\| \leq 1$.

Remark 1.7. As per the definitions in [1], all cross norms satisfy this condition so $\|\cdot\|_\beta$ is a cross norm if and only if $\|\cdot\|_\lambda \leq \|\cdot\|_\beta \leq \|\cdot\|_\gamma$.

Definition 1.8. The *projective cross norm* on $A_1 \otimes A_2$ is denoted by $\|\cdot\|_\gamma$ and defined as

$$\|\xi\| = \inf\left\{\sum_{k=1}^n \|x_{1,k}\| \|x_{2,k}\| : \xi = \sum_{k=1}^n x_{1,k} \otimes x_{2,k}\right\}.$$

Remark 1.9. Let $\|\cdot\|_\beta$ be a cross norm on $A_1 \otimes A_2$. For ξ in $A_1 \otimes A_2$, and $\epsilon > 0$, there exists a representation $\xi = \sum_{k=1}^n x_{1,k} \otimes x_{2,k}$ in $A_1 \otimes A_2$, such that

$$\|\xi\|_\beta \leq \sum_{k=1}^n \|x_{1,k} \otimes x_{2,k}\|_\beta = \sum_{k=1}^n \|x_{1,k}\| \|x_{2,k}\| \leq \|\xi\|_\gamma + \epsilon.$$

So, the projective norm majorises all cross norms. That it is itself a cross norm is evident from the computation

$$\|x_1\| \|x_2\| = \|x_1 \otimes x_2\|_\gamma \leq \|x_1 \otimes x_2\|_\lambda \leq \|x_1\| \|x_2\|.$$

The same computation reveals that all norms ‘between’ the two norms are cross norms. projective norm

1.1 C^* cross-norms

The C^* norms on $A_1 \otimes A_2$ can always be extended to unitisations.

Theorem 1.10. *Let A_1 and A_2 be C^* -algebras with a C^* norm $\|\cdot\|_\beta$ on $A_1 \otimes A_2$. There exists a C^* norm $\|\cdot\|_{\beta_0}$ on $(A_1)_I \otimes (A_2)_I$ which extends $\|\cdot\|_\beta$.*

Proof. Let π be a faithful representation of $A_1 \otimes_\beta A_2$. Then there exist representations π_i of A_i on the same Hilbert Space $\mathcal{H}$ satisfying

$$\pi(x_1 \otimes x_2) = \pi_1(x_1)\pi_2(x_2) = \pi(x_2)\pi_1(x_1)$$

for all x_1 in A_1 , all x_2 in A_2 and $\pi_1(x_1) = \lim_\lambda \pi(x_1 \otimes u_\lambda)$ for an approximate uni $\{u_\lambda\}$ of A_1 . Since π is faithful, $\|x_1 \otimes x_2\|_\beta = \|\pi(x_1 \otimes x_2)\|$. Define $\pi_i^0(x, \lambda) = \pi_i(x_i) + \lambda$ on $(A_i)_I$. Since the ranges of π_1^0 and π_2^0 commute with each other, $\pi_0(x_1 \otimes x_2) = \pi_1^0(x_1)\pi_2^0(x_2)$ defines a valid representation of $(A_1)_I \otimes (A_2)_I$. Define $\|x\|_{\beta_0} = \|\pi_0(x)\|$. This norm extends $\|\cdot\|_\beta$. $\square$

In general, the projective cross-norm is not a C^* norm [2, Remark 4.4]. So we consider

Definition 1.11. The maximal C^* norm defined by

$$\|x\|_{\max} = \sup\{\|\pi(x)\| : \pi \text{ is a representation of } A_1 \otimes A_2\}$$

The max norm is a cross norm.

Remark 1.12. For $\xi = \sum_{k=1}^n x_{1,k} \otimes x_{2,k}$, in $A_1 \otimes A_2$, we have

$$\|\pi(\xi)\| = \left\| \sum_{k=1}^n \pi(x_{1,k} \otimes x_{2,k}) \right\| \leq \sum_{k=1}^n \|\pi(x_{1,k} \otimes x_{2,k})\| \leq \sum_{k=1}^n \|x_{1,k} \otimes x_{2,k}\|.$$

This shows that $\|x\|_{\max} \leq \|x\|_\gamma$.

In addition, for any representation π_1 of A_1 and π_2 of A_2 , we have

$$\begin{aligned} \|\pi_1(x_1)\|\|\pi_2(x_2)\| &= \|\pi_1(x_1)\pi_2(x_2)\| = \|(\pi_1 \otimes \pi_2)(x_1 \otimes x_2)\| \\ &\leq \|x_1 \otimes x_2\|_{\max} \leq \|x_1 \otimes x_2\|_\gamma \leq \|x_1\|\|x_2\|. \end{aligned}$$

By choosing faithful representations π_1 and π_2 , the left hand side and right hand side become equal.

Definition 1.13. The minimal C^* cross-norm $\|\cdot\|_{\min}$ is defined by

$$\|\xi\|_{\min} = \sup\{\|(\pi_1 \otimes \pi_2)(\xi)\| : \begin{array}{l} \pi_1 \text{ is a representation of } A_1, \\ \pi_2 \text{ is a representation of } A_2 \end{array} \}.$$

It is immediate that $\|\cdot\|_{\min} \leq \|\cdot\|_{\max} \leq \|\cdot\|_\gamma$.

Remark 1.14. To see that $\|\cdot\|_{\min}$ is a cross norm, we only need to show that $\|\cdot\|_{\lambda} \leq \|\cdot\|_{\min}$. For f in A_1^* , there exist a representation π_1 of A_1 on a Hilbert Space $\mathcal{H}_1$ and ξ_1, η_1 in $\mathcal{H}_1$ satisfying $\|f\| = \|\xi_1\| \|\eta_1\|$ and $f(a) = \langle \eta_1, \pi_1(a) \xi_1 \rangle$ for all a in A_1 . Similarly, for g in A_2^* , there exist a representation π_2 of A_2 on a Hilbert space $\mathcal{H}_2$ and ξ_2, η_2 in $\mathcal{H}_2$ such that $g(b) = \langle \eta_2, \pi_2(b) \xi_2 \rangle$ for all b in A_2 and $\|g\| = \|\xi_2\| \|\eta_2\|$. This means for $x = \sum_{k=1}^n x_{1,k} \otimes x_{2,k}$ in $A_1 \otimes A_2$, we have

$$\begin{aligned}
& |(f \otimes g)x| \\
&= \left| \sum_{k=1}^n f(x_{1,k})g(x_{2,k}) \right| \\
&= \left| \sum_{k=1}^n \langle \eta_1, \pi_1(x_{1,k}) \xi_1 \rangle \langle \xi_2, \pi_2(x_{2,k}) \xi_2 \rangle \right| \\
&= \left| \sum_{k=1}^n \langle \eta_1, \pi_1(x_{1,k}) \xi_1 \rangle \langle \xi_2, \pi_2(x_{2,k}) \xi_2 \rangle \right| \\
&= \left| \sum_{k=1}^n \langle \eta_1 \otimes \eta_2, (\pi_1 \otimes \pi_2)(x_{1,k} \otimes x_{2,k}) (\xi_1 \otimes \xi_2) \rangle \right| \\
&= |\langle \eta_1 \otimes \eta_2, (\pi_1 \otimes \pi_2)(x) (\xi_1 \otimes \xi_2) \rangle| \\
&\leq \|\eta_1\| \|\eta_2\| \|(\pi_1 \otimes \pi_2)(x)\| \|\xi_1\| \|\xi_2\| \\
&= \|f\| \|g\| \|x\|_{\min}
\end{aligned}$$

In other words, $\|x\|_{\gamma} \leq \|x\|_{\min}$.

Thus, $\|\cdot\|_{\min}$ is a cross norm.

1.2 Universal Property

Theorem 1.15. For representations π_1 , respectively π_2 , of A_1 , respectively A_2 , onto a Hilbert Space $\mathcal{H}$ such that the images of π_1 and π_2 commute, there exists a unique representation π of $A_1 \otimes_{\max} A_2$ on the same space $\mathcal{H}$ such that

$$\pi(x_1 \otimes x_2) = \pi_1(x_1) \pi_2(x_2)$$

and the image of π is the C^* -algebra generated by the images of π_1 and π_2 .

Proof. Define $\pi(x_1 \otimes x_2) = \pi_1(x_1) \pi_2(x_2)$ for x_1 in A_1 , and x_2 in A_2 . For $x = \sum_{k=1}^n x_{1,k} \otimes x_{2,k}$ and $y = \sum_{l=1}^m y_{1,l} \otimes y_{2,l}$ in $A_1 \otimes A_2$, we have

$$\begin{aligned}
& \pi(x_{1,k} \otimes x_{2,k}) \pi(y_{1,l} \otimes y_{2,l}) \\
&= \pi_1(x_{1,k}) \pi_2(x_{2,k}) \pi_1(y_{1,l}) \pi_2(y_{2,l}) \\
&= \pi_1(x_{1,k}) \pi_1(y_{1,l}) \pi_2(x_{2,k}) \pi_2(y_{2,l}) \\
&= \pi_1(x_{1,k} y_{1,l}) \pi_2(x_{2,k} y_{2,l}) \\
&= \pi(x_{1,k} y_{1,l} \otimes x_{2,k} y_{2,l}) \\
&= \pi((x_{1,k} \otimes x_{2,k})(y_{1,l} \otimes y_{2,l}))
\end{aligned}$$

for every k in $\{1, 2, \dots, n\}$ and l in $\{1, 2, \dots, m\}$. Taking linear combinations over k, l we get

$$\pi(x)\pi(y) = \pi(xy).$$

Also, $\|(\pi_1 \otimes \pi_2)(\sum_{k=1}^n x_{1,k} \otimes x_{2,k})\| \leq \|\sum_{k=1}^n x_{1,k} \otimes x_{2,k}\|_{\max}$. So π is norm decreasing. Hence there exists an extension of π to all of $A_1 \otimes_{\max} A_2$.

Moreover, any other representation ρ which satisfies $\rho(x_1 \otimes x_2) = \pi_1(x_1) \otimes \pi_2(x_2)$ must agree with π on the completion because they agree on a dense subset. $\square$

Remark 1.16. This shows that $A_1 \otimes_{\max} A_2$ is embedded inside $A_1 \otimes_{\beta} A_2$ for every C^* cross-norm β .

Theorem 1.17. *Let A_1, A_2 be C^* -algebras and Denote $A_1 \otimes_{\min} A_2$ by $A_{\min}$ and $A_1 \otimes_{\max} A_2$ by $A_{\max}$. Then*

1. *If $\omega_1 \in (A_1^*)_+$, and $\omega_2 \in (A_2^*)_+$, then $\omega = \omega_1 \otimes \omega_2$ is in $((A_{\max}^*)_+$ and induces the representation $(\pi, \mathcal{H}, \xi)$ which is the tensor product of the representations $(\pi_1, \mathcal{H}_1, \xi_1)$ and $(\pi_2, \mathcal{H}_2, \xi_2)$.*

2.

$$\|x\|_{\min} = \sup \left\{ \frac{(\omega_1 \otimes \omega_2)(a^* x^* x a)^{1/2}}{(\omega_1 \otimes \omega_2)(a^* a)^{1/2}} : \begin{array}{l} \omega_1 \in (A_1^*)_+, \\ \omega_2 \in (A_2^*)_+, \\ a \in A_{\min} \end{array} \right\}$$

Proof. Define $\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2$, $\xi = \xi_1 \otimes \xi_2$ and $\pi = \pi_1 \otimes \pi_2$. Then,

$$\begin{aligned} & \langle \xi, \pi(x_1 \otimes x_2) \xi \rangle \\ &= \langle \xi_1 \otimes \xi_2, (\pi_1(x_1) \otimes \pi_2(x_2))(\xi_1 \otimes \xi_2) \rangle \\ &= \omega_1(x_1) \omega_2(x_2) \\ &= (\omega_1 \otimes \omega_2)(x_1 \otimes x_2). \end{aligned}$$

Taking finite linear combinations, we get that $\omega(x) = \langle \xi, \pi(x) \xi \rangle$ for all x in the algebraic tensor product $A_1 \otimes A_2$. This shows that ω is positive in $A_{\max}$. Also, $\pi_1(A_1)\xi_1$ is dense in $\mathcal{H}_1$, $\pi_2(A_2)\xi_2$ is dense in $\mathcal{H}_2$. So their tensor product $(\pi_1(A_1)\xi_1) \otimes (\pi_2(A_2)\xi_2) = \pi(A_1 \otimes A_2)\xi$ is dense in $\mathcal{H}$. Thus, ξ is cyclic under π .

For the second assertion, pick x in $A_{\min}$. Given representations π_1 , respectively π_2 , of A_1 , respectively A_2 , we can express them as direct sums of irreducible representations as $\pi_1 = \oplus_i \pi_{1,i}$ and $\pi_2 = \oplus_j \pi_{2,j}$. Then

$$\|(\pi_1 \otimes \pi_2)(x)\| = \sup_{i,j} \{(\pi_{1,i} \otimes \pi_{2,j})(x)\}.$$

So that

$$\|x\|_{\min} = \sup \|(\pi_1 \otimes \pi_2)(x)\|,$$

where the sup is taken over cyclic representations only. **proof is incomplete.** Let ξ_i be the cyclic vector for π_i , $i = 1, 2$. Replacing π_1 , respectively π_2 with $\omega_1, \in (A_1^*)_+$, respectively $\omega_2 \in (A_2^*)_+$, we get

$$\begin{aligned}
& \|(\pi_1 \otimes \pi_2)(x)\| \\
&= \|(\pi_1 \otimes \pi_2)(x)^*(\pi_1 \otimes \pi_2)(x)\|^{1/2} \\
&= \sup\{\langle \eta, (\pi_1 \otimes \pi_2)(x^*x)\eta \rangle : \eta \neq 0\} \\
&= \sup\left\{\frac{\langle (\pi_1 \otimes \pi_2)(a)\xi, (\pi_1 \otimes \pi_2)(x^*x)(\pi_1 \otimes \pi_2)(a)\xi \rangle}{\langle (\pi_1 \otimes \pi_2)(a)\xi, (\pi_1 \otimes \pi_2)(a)\xi \rangle}\right\} \\
&= \sup\left\{\frac{\langle \xi, (\pi_1 \otimes \pi_2)(a^*x^*xa)\xi \rangle}{\langle \xi, (\pi_1 \otimes \pi_2)(a^*a)\xi \rangle} : a \neq 0, a \in A_1 \otimes A_2\right\} \\
&= \sup\left\{\frac{(\omega_1 \otimes \omega_2)(a^*x^*xa)}{(\omega_1 \otimes \omega_2)(x^*x)} : a \in A_1 \otimes A_2\right\}
\end{aligned}$$

□

Theorem 1.18. *Given C^* -algebras A_1 , and A_2 , the linear functional ω on $A_1 \otimes_{\min} A_2$ is a state if and only if $\Phi'_1(\omega) : A_1 \rightarrow A_2^*$ is completely positive and $\Phi'_1(1) = 1$*

Proof. Say ω is in $(A_1 \otimes_{\min} A_2)^*$ is a state. Now we want to pick an arbitrary positive integer n , and show that $\Phi'_1(\omega)^{(n)}$ is positive. To show this, we take an arbitrary positive element in $A_1^{(n)}$ and show that it is mapped to a positive element. Positive elements in $A_1^{(n)}$ are convex combinations of elements of the form $[x_i^*x_j]$ with $x_1, \dots, x_n$ in A_1 . So, we want to show that $\Phi'_1(\omega)^{(n)}([x_i^*x_j]) = [\Phi'_1(\omega)(x_i^*x_j)]$ is a positive element in $(A_2^*)^{(n)}$. Let us call it Γ . So there is a representation $(\pi, \mathcal{H})$ and vectors ξ, η in $\mathcal{H}$ such that

$$\Gamma((y_j)) = \langle \eta, \pi((y_j))\xi \rangle.$$

□

References

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- [2] Gilles Pisier. *Tensor Products of C^* -Algebras and Operator Spaces: The Connes–Kirchberg Problem*. London Mathematical Society Student Texts. Cambridge University Press, 2020.